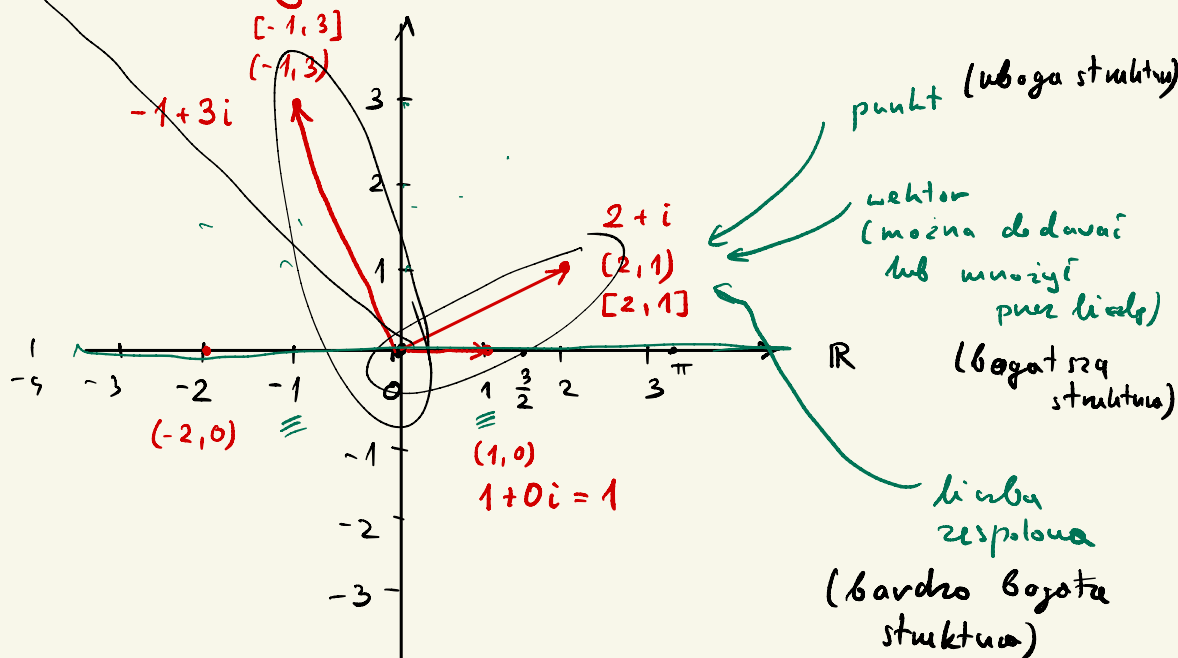
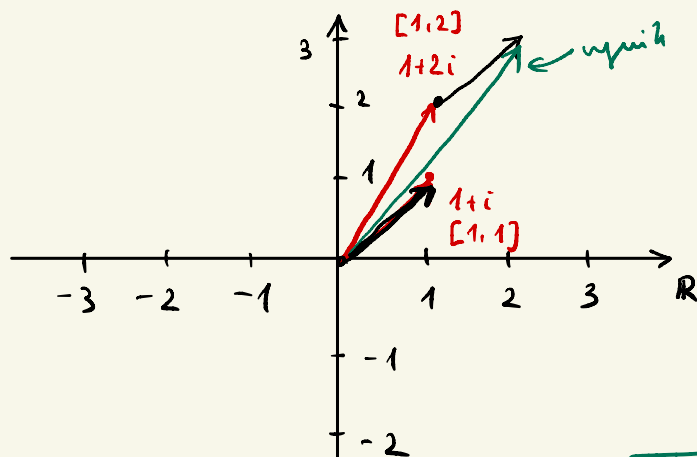




Liczby zespolone



Działania na liczbach zespolonych



$$\begin{aligned} (1+2i) + (1+i) &= \\ &= 2+3i \end{aligned}$$

$$[1, 2] - [1, 1] = [0, 1]$$

$$[1, 2] + [1, 1] = [2, 3]$$

$$\begin{aligned} (1+2i) - (1+i) &= 0+i \\ &= i \end{aligned}$$

$$(3+2i) + (5+i) = 8+3i$$

$$(3+2i) - (5+i) = -2+i$$

$$(-4+i) + (2-2i) = -2-i$$

$$(-4+i) - (2-2i) = -6+3i$$

$$(2+x)(-1+3x) = -2 - x + 6x + 3x^2 = -2 + 5x + 3x^2$$

$$(2+i)(-1+3i) = -2 + 5i + 3i^2 = -5 + 5i$$

$$-2 - i + 6i + 3i^2 \quad \sqrt{-4+5i}$$

Ⓢ
(complex
numbers)

$$i^2 = -1$$

imaginary

(jedinostka urojona)

$$a(b+c) = ab + ac$$

$$\sqrt{-3} = \sqrt{3}i$$

$$\sqrt{-4} = 2i$$

$$(-1+2i) \cdot (3+4i) = -3 - 4i + 6i + 8i^2 =$$

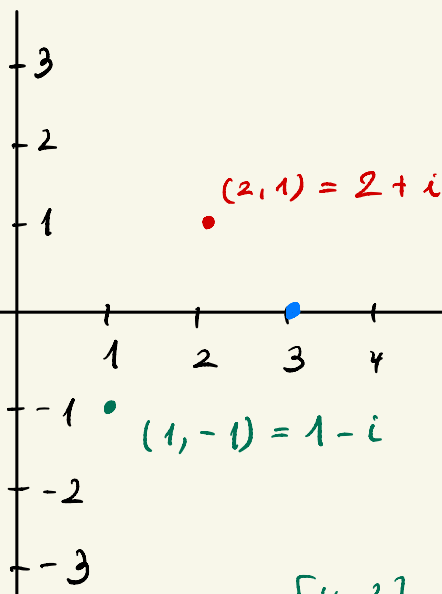
$$= -3 + 2i - 8 = -11 + 2i$$

$$\left[\begin{array}{l} (\text{liczba rzeczywista})^2 \geq 0 \\ (\text{liczba zespolona})^2 \text{ moze byc ujemna} \end{array} \right]$$

Punkty (2bitor)
 Wektory (punkty, które można dodać)

Linie zespolone

(wektory, które można też mnożyć i dzielić) (ciężko)



$$[2, 1] + [1, -1] = [3, 0]$$

$$\frac{[4, 2]}{[2, 2]} = [2, 1]$$

$$(2 + i) + (1 - i) = 3 + 0i = 3$$

$$\frac{[4, 2]}{[2, 0]} = [2, ?]$$

$$(4 + 2i) + (-1 + 3i) = 3 + 5i$$

$$(-2 + i) + (3 + 3i) = 1 + 4i$$

$$(2 + i) \cdot (1 - i) = 2 - 2i + i - \overset{-1}{\boxed{i^2}} = 3 - i$$

$$(2 + x) \cdot (1 - x) = 2 - 2x + x - x^2$$

$$(4 + 2i) \cdot (-1 + 3i) = -4 + 12i - 2i + \overset{-6}{\boxed{6i^2}} = -10 + 10i$$

$$\begin{aligned} (-2+i)(3+3i) &= -6 - 6i + 3i + \boxed{3i^2} \\ &\stackrel{\sim}{=} \stackrel{\sim}{=} \stackrel{\sim}{=} \stackrel{\sim}{=} -6 - 3i - 3 \\ &= -9 - 3i \end{aligned}$$

(1) Jak podzielić liczbę zespoloną przez liczbę rzeczywistą?

$$\frac{4+6i}{2} = 2+3i$$

$$\frac{9+12i}{3} = 3+4i$$

$$\frac{25+10i}{-10} = \frac{25}{-10} + \frac{10i}{10} = -\frac{5}{2} + i$$

$$(2) \quad \frac{z}{w} = \frac{z \cdot v}{w \cdot v}$$

$$\frac{4+2i}{\boxed{(1+i)} \cdot \cancel{v}} \cdot \frac{v}{\cancel{v}} = \frac{(4+2i)}{1+i} \cdot \frac{(1-i)}{1-i} = \frac{6-2i}{2} = 3-i$$

↳ chciałbym znaleźć liczbę v taką, żeby $(1+i) \cdot v$ było l. rzeczyw.

$$\overbrace{(1+i)(1-i)}^{\text{brak } i} = 1 \cdot 1 - \cancel{i} + \cancel{i} - \underbrace{i^2}_{=-1} = 2$$

$$\frac{7+7i}{2+i} \cdot \frac{2-i}{2-i} = \frac{21+7i}{5} = \frac{21}{5} + \frac{7}{5}i$$

$$\frac{\overline{3+i}}{1+2i} \cdot \frac{1-2i}{1-2i} = \frac{3-6i+i-2i^2}{5} = \frac{5-5i}{5} = 1-i$$

$$\frac{4+3i}{1-i} =$$

$$\frac{2-3i}{2+3i} =$$

W dwum

$$z = a + bi$$

liczy rzeczywiste

$$\bar{z} = a - bi$$

(sprzężenie zespolone liczby z)

$$\overline{5+2i} = 5-2i$$

$$\overline{-3-i} = -3+3i$$

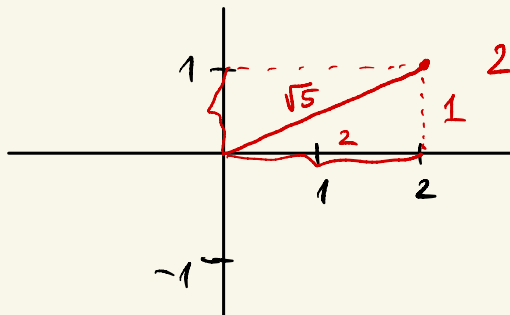
$$\overline{\overline{5+2i}} = \overline{5-2i}$$

$$= 5+2i$$

$$\overline{\bar{z}} = z$$

$z \cdot \bar{z} =$ liczba rzeczywista

$$(a+bi)(a-bi) = a^2 - bi^2 = a^2 + b^2$$



$$2+i = z$$

$$\bar{z} = 2-i$$

$$5 = 2^2 + 1^2$$

$$\underline{z \cdot \bar{z}} = (2+i)(2-i)$$

$$= 2^2 + 1^2$$

$$z \cdot \bar{z} = a^2 + b^2 = (\text{odlegość od zera})^2$$

$$|z| = \text{odlegość } z \text{ od zera} = \text{moduł } z$$

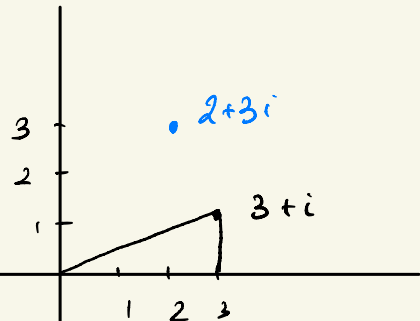
$$z \cdot \bar{z} = |z|^2$$

$$|3+i| = \sqrt{3^2 + 1^2} = \sqrt{10}$$

$$|3+i| = \sqrt{(3+i)(3-i)}$$

$$9 + 3i - 3i + 1$$

" 10



$$|2+3i| = \sqrt{2^2 + 3^2} = \sqrt{13}$$

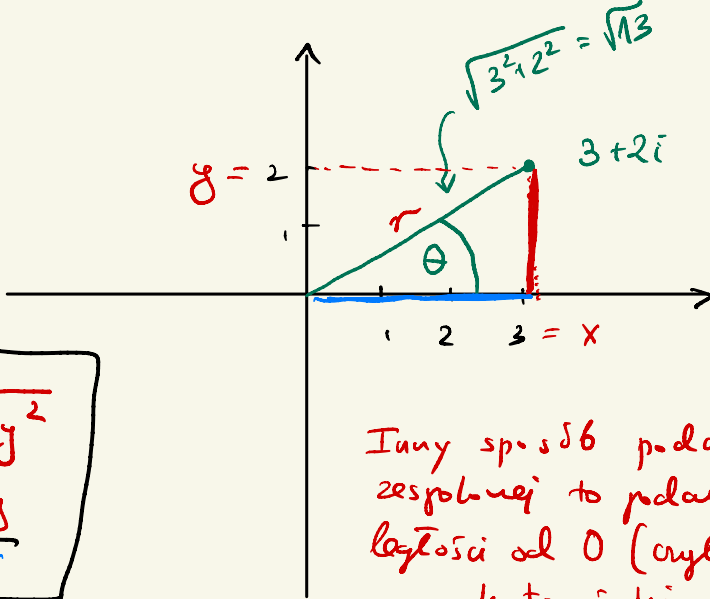
$$|2+3i| = \sqrt{(2+3i)(2-3i)}$$

$2^2 + 3^2$

$$z = \underbrace{a}_{\substack{\text{część} \\ \text{rzeczywista} \\ \text{Re}(z)}} + \underbrace{bi}_{\substack{\text{część} \\ \text{urojona} \\ \text{Im}(z)}}$$

$$\text{Re}(7-3i) = 7$$

$$\text{Im}(7-3i) = -3$$



$$r = \sqrt{x^2 + y^2}$$

$$\tan \theta = \frac{y}{x}$$

Inny sposób podania liczby zespolonej to podanie jej odległości od 0 (czyli modułu) oraz kąta jaki tworzy ona z dodatnią częścią osi OX .

$$\sin \theta = \frac{y}{r} \leadsto y = r \sin \theta$$

$$\cos \theta = \frac{x}{r} \leadsto x = r \cos \theta$$

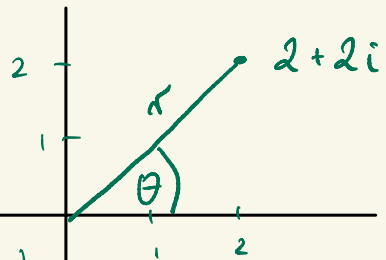
$$\underset{x}{2} + \underset{y}{2}i = \underbrace{\sqrt{8} \cos 45^\circ}_2 + (\sqrt{8} \sin 45^\circ)i$$

$$\sqrt{8} \cdot \cos 45^\circ = \frac{\sqrt{2}}{2} \cdot \sqrt{8} = \frac{\sqrt{16}}{2} = \frac{4}{2} = 2$$

$$r = \sqrt{2^2 + 2^2} = \sqrt{8}$$

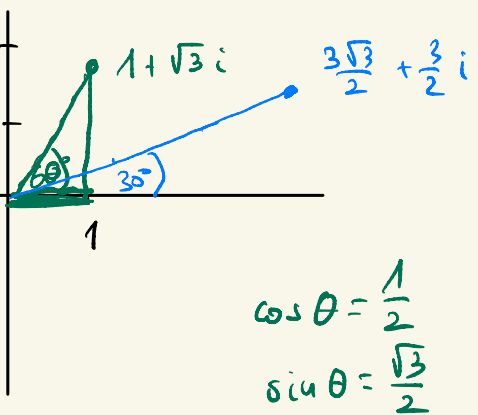
$$\theta = 45^\circ$$

$$\rightarrow \sqrt{8} (\cos 45^\circ + i \sin 45^\circ) = r (\cos \theta + i \sin \theta)$$



$$1 + \sqrt{3}i = 2(\cos \theta + i \sin \theta)$$

$$= 2(\cos 60^\circ + i \sin 60^\circ)$$



$$|1 + \sqrt{3}i| = \sqrt{1^2 + \sqrt{3}^2} = \sqrt{1 + 3}$$

$$= 2$$

$$\cos \theta = \frac{1}{2}$$

$$\sin \theta = \frac{\sqrt{3}}{2}$$

$$z = \frac{3\sqrt{3}}{2} + \frac{3}{2}i = 3(\cos \varphi + i \sin \varphi)$$

$$\sin \varphi = \frac{\frac{3}{2}}{3} = \frac{1}{2}$$

$$|z| = \sqrt{\left(\frac{3\sqrt{3}}{2}\right)^2 + \left(\frac{3}{2}\right)^2} = \sqrt{\frac{9 \cdot 3}{4} + \frac{9}{4}} = \sqrt{\frac{36}{4}} = \frac{6}{2} = 3$$

$$v = 2(\cos 60^\circ + i \sin 60^\circ)$$

$$z = 3(\cos 30^\circ + i \sin 30^\circ)$$

$$\sin(\theta + \varphi) = \sin \theta \cos \varphi + \sin \varphi \cos \theta$$

$$\cos(\theta + \varphi) = \cos \theta \cos \varphi - \sin \theta \sin \varphi$$

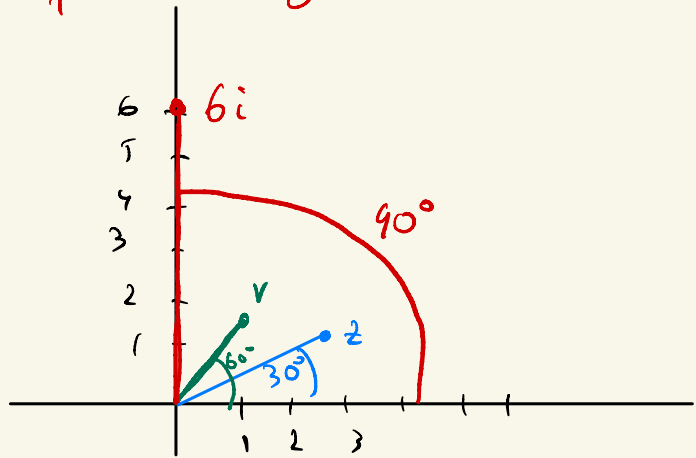
$$v \cdot z = 2 \cdot 3 (\cos 60^\circ \cos 30^\circ + i \cos 60^\circ \sin 30^\circ$$

$$+ i \sin 60^\circ \cos 30^\circ - \sin 60^\circ \sin 30^\circ)$$

$$= 2 \cdot 3 (\cos(60^\circ + 30^\circ) + i \sin(60^\circ + 30^\circ))$$

$$= |v| \cdot |z| (\cos(\theta + \varphi) + i \sin(\theta + \varphi))$$

$$= 6 \left(\underbrace{\cos 90^\circ}_1 + i \underbrace{\sin 90^\circ}_0 \right) = 6i$$



Sprawdzenie:

$$\underbrace{(1 + \sqrt{3}i)} \cdot \underbrace{\left(\frac{3\sqrt{3}}{2} + \frac{3}{2}i \right)} = \cancel{\frac{3\sqrt{3}}{2}} + \frac{3}{2}i + \frac{9}{2}i - \cancel{\frac{3\sqrt{3}}{2}} = \frac{12}{2}i = 6i$$

Jeśli $z = |z| (\cos \theta + i \sin \theta)$

$v = |v| (\cos \varphi + i \sin \varphi)$

to $zv = |z||v| (\cos(\theta + \varphi) + i \sin(\theta + \varphi))$

mnożymy moduły i dodajemy kąty

$$z = \boxed{4 (\cos 12^\circ + i \sin 12^\circ)}$$

$$v = 2 (\cos 20^\circ + i \sin 20^\circ)$$

$$\text{znajdi } z = 8(\cos 32^\circ + i \sin 32^\circ)$$

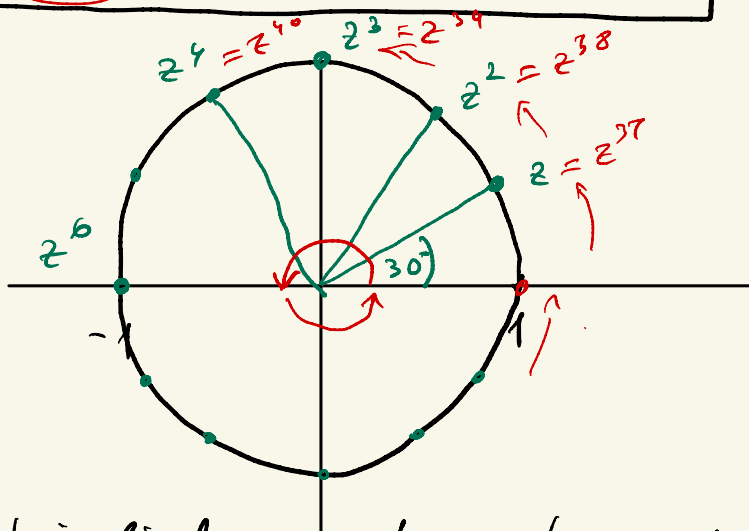
$$z^2 = z \cdot z = 16(\cos 24^\circ + i \sin 24^\circ)$$

$$z^3 = z^2 \cdot z = 64(\cos 36^\circ + i \sin 36^\circ)$$

$$z = |z|(\cos \theta + i \sin \theta)$$

$$z^2 = |z|^2(\cos(2\theta) + i \sin(2\theta))$$

$$z^n = |z|^n (\cos(n\theta) + i \sin(n\theta))$$



wszystkie będą na tym samym okręgu mają ten sam moduł = 1

jeżeli $|z|=1$, to $z^n = \cos(n\theta) + i \sin(n\theta)$

$$z = \cos 30^\circ + i \sin 30^\circ = \frac{\sqrt{3}}{2} + \frac{1}{2}i$$

$$z^2 = \cos 60^\circ + i \sin 60^\circ$$

$$z^3 = \underbrace{\cos 90^\circ}_0 + i \underbrace{\sin 90^\circ}_1 = i$$

$$z^4 = \cos 120^\circ + i \sin 120^\circ$$

$$z^6 = \underbrace{\cos 180^\circ}_{-1} + i \underbrace{\sin 180^\circ}_0 = -1$$

$$z^{12} = \cos 360^\circ + i \sin 360^\circ = 1$$

$$z^{40} = \underbrace{z^{36}}_1 \cdot z^4 = 1 \cdot z^4 = \cos 120^\circ + i \sin 120^\circ = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$$